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Multimodal Fusion and Supervised Contrastive Learning for Robust Document Image Classification

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ABSTRACT

As various industries adapt to the digital transformation, their ability to effectively manage and evaluate vast amounts of digital documents has become essential to operational success. In this regard, automated and accurate classification of such documents plays a critical role. Traditionally, unimodal approaches to document image classification (DIC) show limited robustness when applied to complex and noisy images, constraining accuracy. In parallel, the growing demand for lightweight and resource-efficient models on edge devices and other resource-constrained environments has increased the need for computationally efficient solutions. This study addresses these limitations by proposing two distinct, independent approaches to DIC, designed for different organizational, industrial, and practical needs. The first approach is a multimodal late-fusion framework that combines visual and textual modalities of a document image to improve classification accuracy while reducing cross-modal error propagation. The second approach is an efficiency-oriented, lightweight framework, referred to as supervised contrastive semantic retrieval (SCSR). Instead of employing computationally heavy dense classification layers, this approach integrates supervised contrastive representation learning with semantic similarity retrieval to directly utilize the learned embedding space for final classification, providing a high-performance, resource-efficient alternative for DIC. Experimental results show that the multimodal late-fusion framework outperforms employed unimodal baselines and prior studies, achieving 93.06% and 92.00% accuracy on the RVL-CDIP and Tobacco-3482 datasets, respectively. Further, the alternative lightweight SCSR framework achieves competitive accuracies of 80.76% and 87.89% on the same benchmarks. Both frameworks were also evaluated on RealDoc-16, an independently collected primary dataset, where they achieved strong performance.

1 | Introduction

Every business highly values document images. As these document images hold intelligent assets and valuable operational data, businesses use them for communication, record keeping, and informed decision-making. With most of the world undergoing a digital transformation, there is an urgent need to develop technologies that can efficiently manage, analyze, classify, and derive value from these document images.

In this regard, artificial intelligence (AI) [1] has been an ally for automating document image classification (DIC), benefiting human workers in cases where human efforts face challenges in handling tasks that require great levels of attention, detail, precision, and accuracy over extended periods. DIC focuses on reading, analyzing, and classifying document images like invoices, letters, or reports. Today, the volume of such document images is increasing rapidly, and developing efficient automation systems for DIC has become a necessity [2].

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Recent advancements in DIC have mainly focused on classifying documents based on their visual appearance, textual content that they contain, or a combination of both. Relying only on the visual appearance of the image suffers from a major limitation, where documents belonging to the same class might appear different, while those from different classes can appear remarkably similar [3]. Content-oriented approaches rely on optical character recognition (OCR) to extract the underlying text and subsequently employ natural language processing (NLP) to classify them. However, poor resolution, blurry, or degraded image quality can produce inconsistent OCR outputs. Such outputs can lead to out-of-vocabulary (OOV) noise, which can significantly degrade classification performance. [4]. In addition, by focusing solely on the textual content, these methods ignore the structural information of the document image, losing important spatial and hierarchical context [5].

In order to overcome the limitations of unimodal methods, there has been a significant shift toward multimodal fusion strategies in the field. By combining visual and textual modalities of a document image, fusion architectures leverage the strengths of one type of modality to compensate for weaker or missing information in the other [6]. This shift has been enhanced from simple feature combination methods to advanced vision-language models (VLMs) like LayoutLM and DocFormer [5, 6]. However, despite their impressive performance, these massive multimodal frameworks present significant limitations and impose a large computation burden [7], processing thousands of input tokens per document page and rapidly saturating the context lengths and memory bandwidth of hardware accelerators [8]. Therefore, they are usually not feasible for deployment in real-world settings with resource-constrained environments, such as edge computing, mobile platforms, or the Internet of Medical Things (IoMT). Further, early or intermediate fusion methods remain prone to cascading error propagation if the text extraction is noisy and highly inconsistent [9, 10].

Although significant progress has been achieved in the field [11], existing research primarily emphasizes maximizing classification accuracy by using increasingly complex architectures [12]. Several studies like [13, 14] report that deploying such complex architectures in resource-constrained environments is difficult because of high computational and memory requirements. Very little attention has been given to understand the trade-offs between classification performance, computational efficiency, and resource requirements.

To address this gap, this study proposes two distinct and independent DIC frameworks, allowing practitioners to select the architecture that best aligns with their specific requirements, preferences, available resources, and latency requirements.

In contrast to existing studies that mainly focus on employing increasingly complex architectures without considering the expense of computational feasibility, this study directly addresses differing operational needs. The first framework, a multimodal fusion approach, utilizes a lightweight two-stream late-fusion architecture to decouple feature extraction from fusion, enhancing modularity and reducing cross-modal error propagation. The second framework, supervised contrastive semantic retrieval (SCSR), relies solely on the textual modality. It removes the dependence on deep classifier layers by combining supervised

contrastive representation learning with semantic similarity retrieval, directly utilizing the learned embedding space for classification.

1.1 | Contributions

The primary contributions of this study can be summarized as follows:

A two-stream late-fusion multimodal classification framework was developed to decouple feature extraction from fusion, enhancing classification accuracy while reducing cross-modal error propagation.

Another lightweight SCSR framework was proposed to combine contrastive representation learning with semantic similarity search, addressing the growing need for low-latency and resource-efficient document classification in resource-constrained environments.

A dedicated cross-domain document image dataset was curated to separately evaluate effectiveness of both proposed approaches beyond standard benchmark datasets.

This study is structured into five sections. Section 2 provides a review of the relevant literature on structure-oriented, content-oriented, and multimodal strategies for DIC. Section 3 introduces the architecture, methodology, and training procedures of the proposed frameworks. Section 4 presents a detailed evaluation of the proposed framework's performance and compares the results with existing techniques. Finally, the conclusion of the work and directions for future research are discussed in Section 5.

2 | Related Work

The task of classifying document images has a rich history, and its importance continues to rise as the world embraces digitization. For the task of efficient organizing, easy assessing, retrieving, and effective indexing, the textual information within a document image is as crucial as the image itself. Accurate classification of document images is essential for such tasks. DIC has been addressed through numerous different approaches, which can broadly be grouped into three main categories: (1) structure/layout-oriented, (2) content-oriented, and (3) multimodal methods. Structure-oriented classification relies on the visual features of the document image, while content-oriented approaches utilize the text extracted from those images for classification. Multimodal methods typically harness the advantages of both structure-oriented and content-oriented approaches. An overview of leading-edge research in the field of DIC is provided in this section, with emphasis on structure/layout-oriented, content-oriented, and multimodal approaches.

2.1 | Structure/Layout-Oriented Approaches

A popular technique is to classify the document images using deep learning (DL) approaches, which makes it possible to extract robust visual features from the structural layout of document images. From a structure/layout-focused standpoint, a prevalent DL approach utilized in image classification is training an artificial neural network (ANN) [15]. However, this training process can be quite lengthy and costly, and other methods and strategies were subsequently proposed. Kang et al. [16] were pioneers in using a convolutional neural network

(CNN) which consisted of 4 layers for classifying document images and achieved a notable classification accuracy of 65.37% on the Tobacco3482 dataset [17]. Deep convolutional neural networks (DCNNs) were applied by Afzal et al. [18] for DIC, highlighting the importance of structural features extracted from the images for precise classification. Subsequently, Tensmeyer and Martinez [19] demonstrated that implementing strategies like preprocessing, augmenting the dataset or leveraging transfer learning approaches, and applying domain adaptation techniques enhanced the performance of a CNN in classifying document images [20]. Yaman et al. [21] fine-tuned several DCNNs such as GoogLeNet, AlexNet, and VGG-16, with findings indicating that VGG-16 provided the best classification performance for document images. An ensemble approach was introduced by Roy et al. [22], in which DCNNs were trained on entire document images and specific regions, and final predictions were made using a support vector machine (SVM) classifier. Making use of run-length histogram (RL) and Fisher vectors (FVs), Csurka [23] extracted shallow features from document images which were then classified using traditional machine learning algorithms. Unique among other studies, Kolsch et al.'s study [24] carried out DIC in real time. A stack generalization-based ensemble of VGG16 models was introduced by Das et al. [25], where each model was trained on document image regions. It was demonstrated by Siddiqui et al. [26] that self-supervised representation learning approaches can generate effective document image representations, especially under conditions where labeled data are limited. Furthermore, Liu et al. [27] conducted two classification tasks and concluded that preprocessing enhanced shallow CNNs outperformed the performance of other deeper CNN models for the classification tasks.

2.2 | Content-Oriented Approaches

From a content-oriented perspective, the classification of document images relies on the textual information contained within the image. A neural network was introduced by Yang et al. [28] to extract semantic insights from document images, utilizing word embeddings derived from pretrained natural language models. In addition to visual tasks, CNNs have been employed in the areas of text classification and comprehension as well. A simple CNN structure featuring a single convolutional layer was recommended by Kim et al. [29], which performed effectively on challenging text classification tasks. Moreover, the adoption of text-focused DL frameworks such as DocBERT [30] and BERT [31] has been utilized for DIC by processing the extracted textual content within the document images. Johnson et al. [32] explored CNNs for text categorization by utilizing word order to improve prediction accuracy. Soni et al. [33] proposed TextConvoNet, a CNN-based framework for binary and multiclass text classification. Additionally, an extensive overview of the various text classification techniques was carried out by Kowsari et al. [34]. The development and rise of learned word-embedding models, such as Word2Vec [35], ELMo [36], FastText [37], and XLNet [38], has also contributed to significant advancements in this area.

2.3 | Multimodal Approaches

Modern methods for DIC often rely on multimodal strategies, which involve combining both textual and visual modalities of the document image. Yang et al. [28] introduced a multimodal

framework for DIC, integrating visual information of the image with text embeddings obtained from pretrained NLP models. Subsequently, Noce et al. [39] enhanced classification accuracy by representing the OCR-extracted text from the image as colored boxes, which were then processed through a CNN. Audebert et al. [3] implemented a multimodal network for DIC, integrating FastText embeddings to extract textual features alongside MobileNetV2 for visual feature extraction. Another multimodal framework was introduced by Lai et al. [40] that combined an RNN to capture textual features and a CNN to extract the visual features of the document image for classification. Further, Jain and Wigington's [41] study compared spatial and feature fusion techniques for multimodal classification, where feature fusion performed better than spatial fusion in most experiments. Bakkali et al. [42] proposed a DL architecture employing a dual-headed architecture, with one head capturing the structural and visual features of the document images through NASNetLarge, while the other head extracted the textual features using BERT. Similarly, numerous other multimodal approaches have been suggested, including those by Ferrando et al. [43], Tyler et al. [44], Cosma et al. [45], and Bakkali et al. [46]. These studies employ various architectural methods to merge both textual and visual features of document images. Some studies adopted late fusion techniques, combining features at the decision-making stage, while others explored early-fusion frameworks that engage each modality early in the process, and additional studies developed end-to-end multimodal frameworks that learn jointly from both input types throughout the entire process.

2.4 | Emerging Paradigms

Recent advances in document understanding have shifted toward VLMs, multimodal large language models (MLLMs), large-scale multimodal pretraining, and advanced representation learning techniques [47, 48]. With their increasing ability to jointly model both textual and visual information in an efficient manner, these frameworks have started to be adapted to document-centric tasks. However, a study by the authors in [49] reports that generic multimodal models underperform on document understanding tasks unless they are explicitly adapted to capture fine-grained document structures. Consequently, models such as DocOwl 1.5 [49] and TextMonkey [50] proposed OCR-free, layout-aware frameworks, which involve token-level text modeling, text grounding, and structural attention to preserve complex spatial hierarchies.

Apart from these developments, contrastive representation learning serves an alternative approach to traditional supervised classification. Unlike traditional frameworks that make direct predictions about probability of a class through dense classification layers, modern approaches make use of embedding spaces where similar multimodal documents naturally cluster [51]. Studies like ColMate [52] and Evo-Retriever [53] demonstrate that including document-specific structural biases through masked contrastive learning, multiview alignment, and adaptive hard-negative mining significantly enhances retrieval and out-of-domain generalization.

3 | Research Methodology

To address the limitations of traditional unimodal approaches in classifying document images and the growing demand for

lightweight and resource-efficient models [54], this study proposes and evaluates two distinct DIC frameworks. Each framework is tailored to different efficiency, resource, and latency requirements. First, it proposes a multimodal fusion framework that leverages the strengths of both visual and textual modalities of the document image and performs the final classification through late fusion. Secondly, an alternative lightweight and efficient SCSR framework is proposed, which relies solely on the textual modality, where the text that is extracted from the document image is converted into dense vector representations through supervised contrastive learning, and final classification is done directly using semantic similarity search within a learned embedding space. Both frameworks are developed and evaluated separately in the same study to prove and compare their complementary strengths in solving different practical problems with respect to different resources and environments. The multimodal fusion framework prioritizes on achieving the highest possible classification accuracy, although with a higher computational complexity and heavier model architecture. In contrast, the SCSR framework is lightweight and is designed on principles of computational efficiency and deployment simplicity, making it more suitable for resource-constrained environments.

3.1 | Approach 1: Multimodal Fusion Framework

Though large-scale architectures such as Layoutlmv3 [55], DocFormer [56], and DocLLM [57] have shown promising results, and they may lead to substantial computational overhead, increased memory requirements, and deployment challenges in resource-constrained environments [55, 58]. In addition, tightly coupled multimodal architectures are prone to cross-modal error propagation even when one modality is noisy or unreliable [9, 50].

To overcome this, the proposed multimodal fusion framework employs a lightweight two-stream late-fusion architecture. By decoupling feature extraction from fusion, the framework improves modularity, reduces cross-modal error propagation, and provides robustness when either modality is degraded [10, 59]. This framework employs a two stream architecture consisting of the following: (i) a visual stream, which considers visual modality of the document image to extract visual features and (ii) a textual stream, which considers the textual modality of the document image to extract textual features. Each stream extracts features of its respective modality independently, and the outputs from both streams are later combined using a late-fusion strategy. The complete architecture of this framework is given in Figure 1.

3.1.1 | Visual Stream

The visual stream leverages the visual modality of the document image and extracts the visual features from document images. This stream utilizes EfficientNetB3 [60], a lightweight and efficient architecture with fewer parameters than other vision-based DL architectures. Employing EfficientNetB3 provides a strong balance between computational cost and classification accuracy. A training subset of the RVL-CDIP dataset was employed to fine-tune the model. Further, a dropout layer was included to address potential overfitting. The fine-tuned model was subsequently employed to perform classification on document images within the held-out test sets for evaluation. The features extracted from

each image in the test sets were input to the fully connected layer of EfficientNetB3, followed by a softmax activation function for final classification. The basic architecture of EfficientNetB3 is given in Figure 2. Algorithm 1 offers an in-depth explanation of the sequential procedure for the visual stream.

3.1.2 | Textual Stream

The textual stream leverages the textual modality of the document image. First, the text within the image is extracted using an OCR engine. Because some document images in the dataset may have low resolution or degraded quality, the output from the OCR may include OOV words, which might consist of missing, unclear, or incorrectly formed words. To overcome this, the extracted text undergoes data preprocessing. Additionally, employing BERT_{base} effectively addresses the effects of OOV words through WordPiece tokenization [31]. The training set of RVL-CDIP dataset, where the text was extracted from the original document images, was employed for fine-tuning the model. The fine-tuned model was subsequently used to classify the extracted text instances within held-out test sets for evaluation. The features extracted from each text instance in the test sets were input to the fully connected layer of BERT_{base}, followed by a softmax activation function for final classification. The detailed procedure of the textual stream workflow is outlined in Algorithm 2.

3.1.3 | Fusion of Visual and Textual Stream

The final prediction of the document image is derived by integrating predictions from both the visual and textual streams, often referred to as multimodal fusion. As observed in prior works [61–63], fusion from multiple modalities is beneficial in terms of a greater reliability and gain of additional information, which enables tasks to be accomplished even if some needed features from a single modality are unreliable, redundant, or absent. With regard to DIC, the visual and textual modalities of the document image are complementary to each other. Some document classes may have an effective visual representation, and in such cases, a visual stream employing a vision-based DL model can achieve higher classification accuracy, while other document images may have a better textual representation and can achieve higher classification accuracy by classifying the text present within a document image with the textual stream. This scenario is better illustrated in Table 1.

Thus, relying on a single modality may lead to inconsistent performance. To address this limitation, a multimodal late fusion framework is employed, which leverages both textual and visual modality of the document image. In this work, the proposed late fusion method utilizes unweighted average ensembling [64], where the logits, i.e., the raw, unnormalized prediction scores produced by each stream prior to softmax activation, are averaged to generate the final output distribution. The approach was evaluated across the “RVL-CDIP” test set, “Tobacco3482,” and “RealDoc-16” dataset. Algorithm 3 offers an in-depth explanation of the sequential procedure for multimodal late fusion.

3.2 | Approach 2: Supervised Contrastive Learning and Semantic Similarity Search

Motivated by the strict hardware limitations in resource-constrained environments [54], an alternative lightweight approach is proposed, referred to as the SCSR framework in the

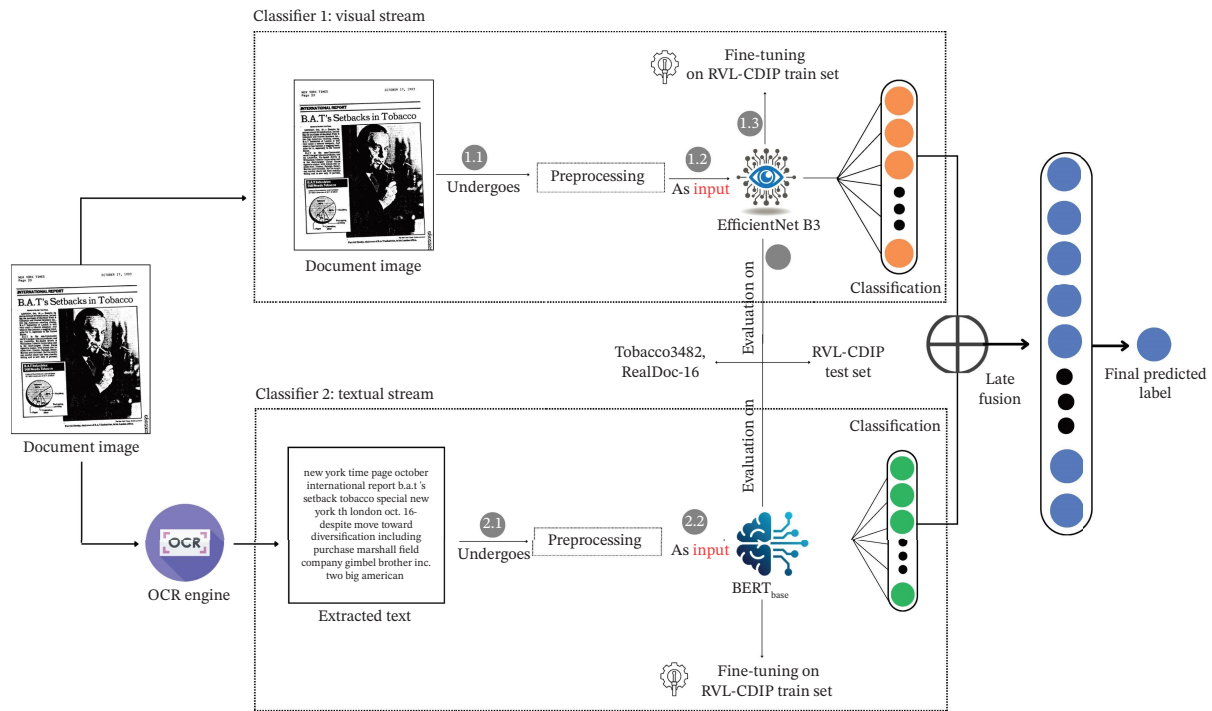


FIGURE 1 | Overall architecture of the proposed multimodal fusion framework.

ALGORITHM 1 | Working of the visual stream.

Input: Document image $\leftarrow I$

Output: Predicted label $\leftarrow \hat{y}_v$

Fine-tuning phase

1. Load the train split of “RVL-CDIP” document image dataset #consists of 3,19,625 instances
2. **def** preprocess_image(I):
 $\text{preprocess_images}(I) \leftarrow I_{\text{pre}}$
3. Initialize pre-trained “EfficientNetB3” model $\leftarrow E$
4. Fine-tune the pre-trained EfficientNetB3 on pre-processed images I_{pre} :
 $\text{fine_tune}(E, I_{\text{pre}}) \leftarrow E_{\text{fine-tuned}}$

Testing phase

5. Load the “RVL-CDIP” test split and “Tobacco3482” dataset
6. Preprocess the document images using Step-2
7. Extract the features of the document image and generate its feature representation:
def feature_extraction ($E_{\text{fine-tuned}}, I_{\text{pre}}$):
 $\text{feature_extraction}(E_{\text{fine-tuned}}, I_{\text{pre}}) \leftarrow F_I$
8. The feature vector F_I (from Step-7) is fed to the model’s fully connected layer, after which a softmax activation is applied to obtain the predicted label $\hat{y}_v$
9. Repeat Steps 7–8 for all instances of the “RVL-CDIP” test set and “Tobacco3482” dataset
10. **return** $\hat{y}_v$

remainder of this study. While self-supervised contrastive learning has made notable advancements in areas like computer vision [65, 66], speech [67, 68], and NLP [69, 70], existing studies

primarily apply contrastive loss to improve embedding representations, while relying on a conventional classifier head for final classification [26, 71]. By removing the dependence on deep

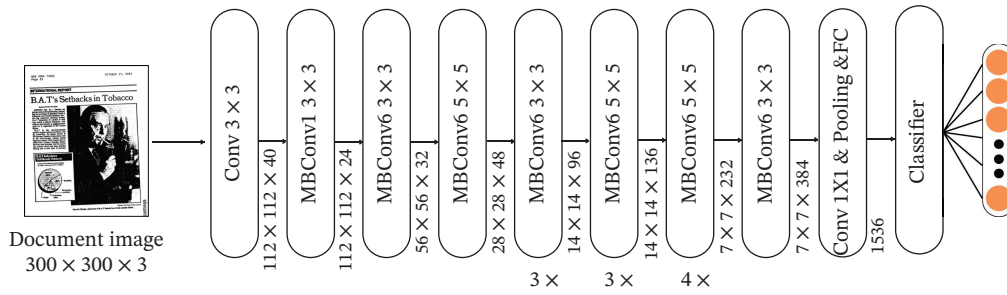


FIGURE 2 | Basic architecture of EfficientNetB3 where “x” represents the number of corresponding layers.

classifier layers, the proposed framework combines supervised contrastive representation learning with semantic similarity retrieval, directly utilizing the learned embedding space for classification. The integration of contrastive learning and retrieval-based classification for DIC remains largely unexplored. As far as we know, current studies in the field have not explored supervised contrastive representation learning and semantic retrieval as a unified classification pipeline.

This approach relies solely on the textual modality, where the text that is extracted from the document image is converted into

dense vector representations through supervised contrastive learning. Subsequently, classification is done directly using semantic similarity search within a learned embedding space. This approach is structured into three major steps: (1) fine-tuning a pretrained embedding model using supervised contrastive learning, (2) storing these vector representations in a retrieval database, and (3) classifying test instances by retrieving their most similar vector representations from the database. Figure 3 illustrates the complete working of this framework.

ALGORITHM 2 | Working of the textual stream.

Input: Document image $\leftarrow \mathbf{I}$

Output: Predicted label $\leftarrow \hat{y}_t$

1. Extract textual content from document images by employing an OCR engine
2. Load the document image datasets. # *RVLCDIP train/val/test, Tobacco3482 and RealDoc-16*
3. **def** ocr_extraction($\mathbf{I}$):
 $\text{ocr_extraction}(\mathbf{I}) \leftarrow \mathbf{T}_{\text{raw}}$ # employ OCR engine to extract text from the document images
4. **def** preprocess_text ($\mathbf{T}_{\text{raw}}$):
 $\text{preprocess_text}(\mathbf{T}_{\text{raw}}) \leftarrow \mathbf{T}_{\text{clean}}$ # apply text preprocessing techniques on raw extracted texts

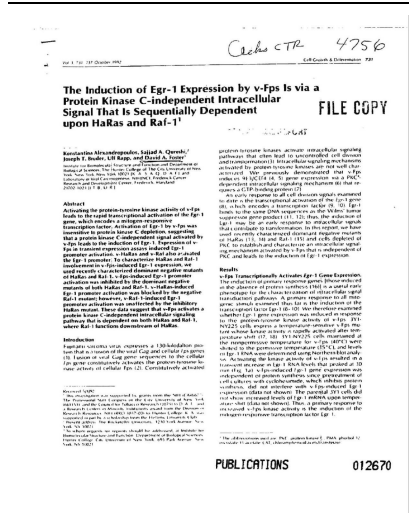
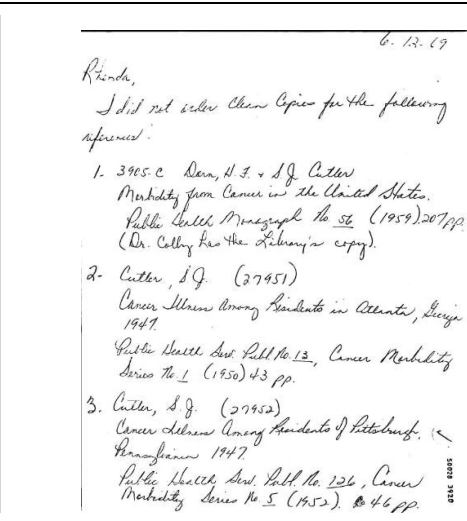
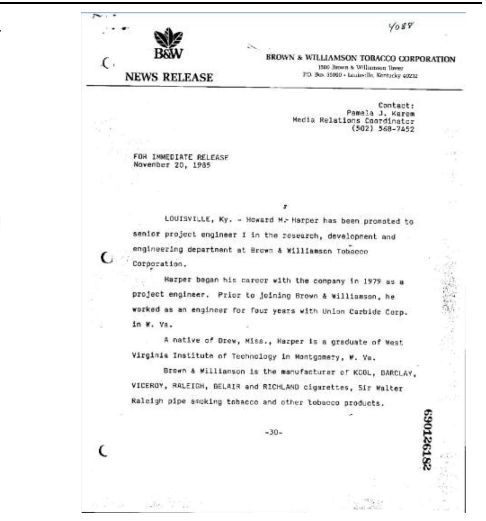
Fine-tuning Phase

5. Initialize pre-trained model: $\mathbf{BERT}_{\text{base}} \leftarrow \mathbf{B}$
6. Load the extracted text set from train split of RVL-CDIP dataset $\leftarrow \mathbf{RVL} - \mathbf{CDIP}_{\text{train}}(\text{OCR})$
7. Fine-tune the pre-trained $\mathbf{BERT}_{\text{base}}$ on $\mathbf{RVL} - \mathbf{CDIP}_{\text{train}}(\text{OCR})$:
 $\text{fine_tune}(\mathbf{B}, \mathbf{RVL} - \mathbf{CDIP}_{\text{train}}(\text{OCR})) \leftarrow \mathbf{B}_{\text{fine-tuned}}$

Testing phase

8. Load the extracted text sets for the “RVL-CDIP test”, “Tobacco3482” and “RealDoc-16” datasets.
9. Preprocess the datasets using Step-4
10. Extract the features of the text and generate its feature representation:
def feature_extraction ($\mathbf{B}_{\text{fine-tuned}}, \mathbf{T}_{\text{clean}}$):
 $\text{feature_extraction}(\mathbf{B}_{\text{fine-tuned}}, \mathbf{T}_{\text{clean}}) \leftarrow \mathbf{F}_T$
11. The feature vector $\mathbf{F}_T$ (from Step-10) is fed to the model’s fully connected layer, after which a softmax activation is applied to obtain the predicted label $\hat{y}_T$
12. Repeat Steps 10–11 for all instances of the “RVL-CDIP” test and “Tobacco3482” dataset
13. **return** $\hat{y}_t$

TABLE 1 | Comparison of visual, textual, and multimodal fusion on sample RVL-CDIP document images.

		
True label: scientific pub	True label: handwritten	True label: presentation
Visual stream: scientific rep	Visual stream: handwritten	Visual stream: letter
Textual stream: scientific pub	Textual stream: presentation	Textual stream: presentation
Multimodal fusion: scientific pub	Multimodal fusion: handwritten	Multimodal fusion: presentation

3.2.1 | Fine-Tuning a Pretrained Embedding Model Using Supervised Contrastive Learning

Each document image in the RVL-CDIP training set was processed by encoding its extracted text with the “bge-small-en” embedding model, introduced by the Beijing Academy of Artificial Intelligence (BAAI) [72]. This embedding model was fine-tuned by applying supervised contrastive learning. For this, an

anchor-positive dataset was constructed using the extracted text instances from the RVL-CDIP training subset, where each instance consisted of the following: (i) an anchor text instance and (ii) a positive text instance, belonging to the same class as of the anchor. The positive instances were selected randomly within their respective classes. By employing the RVL-CDIP training dataset, 319,625 pairs of anchor-positive text samples were

ALGORITHM 3 | Working of the multimodal late fusion.

Input: Predicted Labels from the visual stream $\hat{y}_v$ and textual stream $\hat{y}_t$

Output: Final fused prediction $\leftarrow \hat{y}_f$

1. Load the “RVL-CDIP” test split, “Tobacco3482” and “RealDoc-16” dataset
2. Collect predictions from visual stream across all m test document images
3. `def predictions_from_visual_stream():`
$$y_v = [y_{v1}, y_{v2}, y_{v3}, y_{v4}, \dots, y_{vm}] \# \text{ predictions from visual stream across } m \text{ test images}$$
4. Collect predictions from textual stream across all m test extracted text instances
5. `def predictions_from_textual_stream():`
$$y_t = [y_{t1}, y_{t2}, y_{t3}, y_{t4}, \dots, y_{tm}] \# \text{ predictions from textual stream across } m \text{ extracted text instances}$$
6. Perform late fusion by averaging logits from both streams
7. `def late_fusion( $y_v, y_t$ ):`
$$y_f = (y_v + y_t) / 2 \# \text{ apply unweighted average ensembling as a late fusion strategy}$$
8. Determine the final class label by selecting the class with maximum fused probability
9. `def classification( $y_f$ ):`
$$y = \operatorname{argmax}_C(y_f) \# \text{ Select the class } C \text{ with maximum fused probability}$$
10. **return** y
11. Repeat Step 6–10 for all instances in the testing datasets

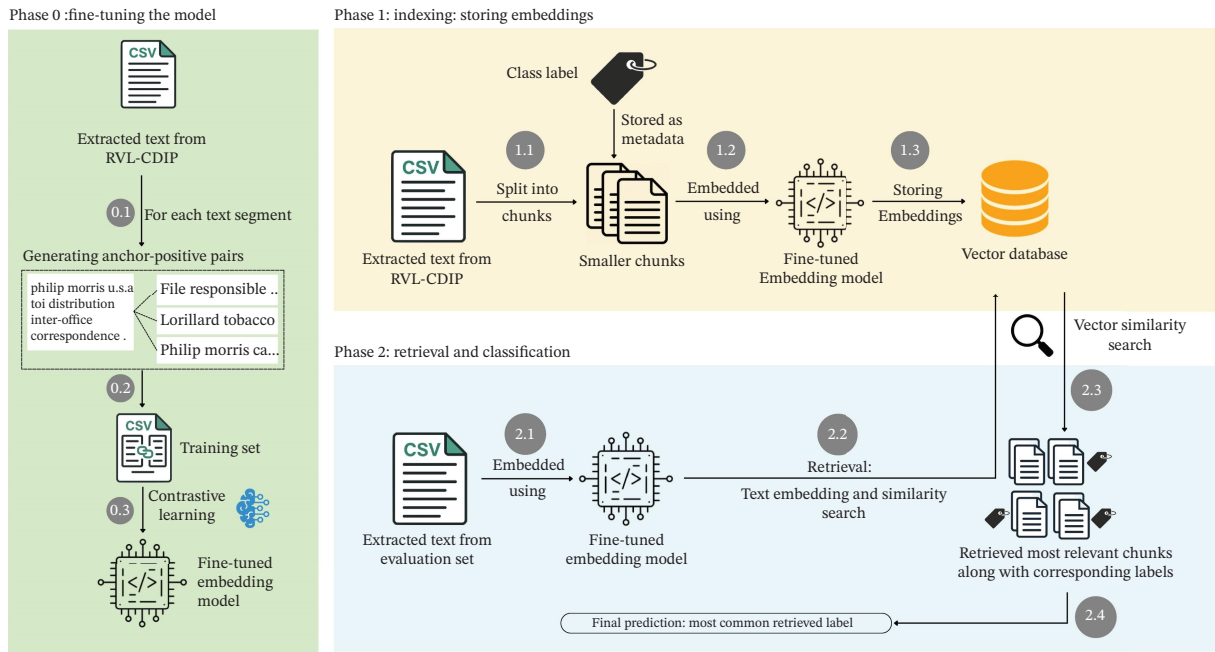


FIGURE 3 | Overall architecture of the proposed supervised contrastive semantic retrieval SCSR framework.

created. This enabled the embedding model to learn embeddings in such a way that text instances from the same class are closely grouped, while those from different classes are spaced apart in the embedding space. This process is often referred to as supervised contrastive learning. Algorithm 4 outlines the sequential workflow of this phase.

3.2.2 | Indexing: Storing Dense Vector Representations for Retrieval

The fine-tuned embedding model i.e., “bge-small-en,” was then employed to generate dense vector representations of the extracted text instances from the RVL-CDIP training dataset. Since “bge-small-en” truncated the sequence at an input of 512

ALGORITHM 4 | Anchor-positive dataset generation and fine-tuning the embedding model.

Input: Pre-processed extracted text from the RVL-CDIP training set

Output: Fine-tuned embedding model

1. Initialize the pre-trained “BAAI/bge-small-en” model $\leftarrow E$
2. Load the “RVL-CDIP” training dataset $\leftarrow D$ # consists of 319,625 text instances
3. Group all text instances by their labels:
for each $(\text{text}_i, \text{label}_i)$ in D do:
Append (text_i) to $\text{label_groups}[\text{label}_i]$
4. Initialize an empty list $T \leftarrow []$
5. for each label in label_groups do:
6. for each text_i in $\text{label_groups}[\text{label}]$ do:
7. # Randomly select one positive text_j from $\text{label_groups}[\text{label}]$, $j \neq i$
8. # Add anchor-positive pair $(\text{text}_i, \text{text}_j)$ to T
9. Repeat Step 5–8 until anchor-positive pairs for every text instance are formed # for 319,625 text instances
10. Construct a training dataset from T
11. Fine-tune the embedding model E using supervised contrastive learning:
Minimize distance between (anchor text_i , positive text_j)
Maximize distance between (anchor text_i , negative text_k)
12. Save the fine-tuned embedding model

tokens, any text instances that were longer than 512 tokens risked truncation, potentially erasing important information. To address this issue, a recursive character-based text splitting strategy was applied, where each text instance was recursively divided into smaller chunks of 512 characters, with consecutive chunks overlapping by 50 characters. This assured that longer text instances in the dataset were split into smaller chunks whenever they exceeded the model's maximum input limit, while preserving contextual continuity between adjacent chunks. This process is often referred to as chunking. This allowed the embedding model to process the complete textual content without information loss. Each chunk, along with its corresponding class label, was then converted into dense vector representations using the fine-tuned model. For effective storage retrieval, the embeddings were indexed and stored using the Facebook AI Similarity Search (FAISS) [73] vector database, which provides GPU-accelerated similarity search, enabling fast nearest-neighbor queries in high-dimensional spaces. This process of storing dense vector representations to a vector database is referred to as indexing. The resulting FAISS vector store contained 1,030,531 dense embeddings, each of dimensionality 384. Algorithm 5 details the stepwise procedure for this phase.

3.2.3 | Classifying Test Instances by Retrieving From the Vector Database

For classification purposes, each text instance from the held-out test sets i.e., RVL-CDIP test set, Tobacco3482 dataset, and RealDoc-16 dataset, was converted into dense vector representations using the fine-tuned embedding model. The resulting vector representation of each instance was then matched against the preexisting dense vectors stored within the vector database. For each text instance within the test sets, 10 chunks with highest semantic similarity were fetched from the vector database. This retrieval was performed using cosine

similarity. Since the retrieved chunks were stored with their class labels during the indexing phase as mentioned in Section 3.2.2, the final prediction for each test instance was decided by conducting a majority vote across the class labels of the retrieved chunks. This process was repeated and evaluated for all the test instances in the RVL-CDIP test set, Tobacco3482 dataset, and RealDoc-16 dataset. The sequential procedure for this phase is outlined in Algorithm 6.

This approach for classifying document images is referred to as **SCSR** in the remainder of this study.

3.3 | Datasets

To evaluate the proposed approaches under different scenarios, three datasets were used: (a) RVL-CDIP [74], (b) Tobacco3482 [75], and (c) an independently gathered dataset, referred to as RealDoc-16. Recent studies show that document classification frameworks achieving strong benchmark performance can experience performance degradations of approximately 15%–30% under distribution shifts and real-world document variations [76]. Therefore, this multidataset strategy was adapted as evaluating only on a single benchmark dataset may not accurately reflect the practical deployment performance. Table 2 depicts the statistical overview of the datasets.

- a. **RVL-CDIP**: RVL-CDIP stands for Ryerson Vision Lab Complex Document Information Processing. Since its introduction by Harley et al. [49], the RVL-CDIP corpus has been regarded as a standard dataset for document classification tasks. This dataset comes with a predefined split for training, validation, and testing and contains document images with substantial layout diversity, scanner artifacts, and visually similar document classes, making it a challenging benchmark for large-scale document classification.

ALGORITHM 5 | Indexing: storing dense vector representations to a vector database.

Input: Pre-processed extracted text from the RVL-CDIP training set

Output: Vector Database with dense vector representations

1. Initialize the fine-tuned “bge-small-en” model $\leftarrow \mathbf{M}$
2. Load the extracted texts from “RVL-CDIP” training dataset $\leftarrow \mathbf{D}$ # consists of 319,625 text instances
3. Initialize an empty list $\mathbf{C} \leftarrow []$ # to store the chunked text instances
4. for each text instance t in $\mathbf{D}$ do:
 - if $\text{length}(t) > 512$ token then:
 - # split t into overlapping chunks of 512 characters with 50-character overlap
 - else:
 - # keep t as a single chunk
 - Append all the resulting chunks to $\mathbf{C}$, also preserving their class labels
5. for each chunk c in $\mathbf{C}$ do:
 - # Generate dense vector representations $v_c \leftarrow \mathbf{M}(c)$
 - # Store (v_c, label_c) into FAISS vector database $\leftarrow \mathbf{VecDB}$
6. After processing all chunks, finalize the FAISS vector database for efficient similarity search
7. Save the Vector Database

ALGORITHM 6 | Classifying test instances by retrieving from the vector database

Input: Pre-processed extracted text from the RVL-CDIP test set (**D1**), Tobacco-3482 (**D2**) and RealDoc-16 (**D3**)

Output: Predicted Label

1. Initialize the fine-tuned “BAAI/bge-small-en” model $\leftarrow \mathbf{M}$
2. Load the “RVL-CDIP” test dataset $\leftarrow \mathbf{D1}$ # consists of 40,000 text instances
3. Load the “Tobacco-3482” dataset $\leftarrow \mathbf{D2}$ # consists of 3482 text instances
4. Load the “RealDoc-16” dataset $\leftarrow \mathbf{D3}$ # consists of 160 text instances
- # Step A: classification on RVL-CDIP test set $\leftarrow \mathbf{D1}$
5. for each text instance $\mathbf{t}$ in $\mathbf{D1}$ do:
6. $\mathbf{v_t} \leftarrow \mathbf{M(t)}$ # generate dense vector representation using the fine-tuned embedding model
7. $\mathbf{N_t} \leftarrow \text{retrieve_top_k}(\mathbf{VecDB}, \mathbf{v_t}, k = 10)$ # retrieve 10 most semantically similar chunks
8. $\mathbf{L_t} \leftarrow \{\text{label}(c) | c \in \mathbf{N_t}\}$ # extract the labels of retrieved chunks
9. $\hat{\mathbf{y}} \leftarrow \text{majority_vote}(\mathbf{L_t})$ # determine the predicted label of the text instance
10. # store $\hat{\mathbf{y}}$ as the predicted label of $\mathbf{t}$
11. After processing all instances in $\mathbf{D1}$, compute evaluation metrics
- # Step B: classification on Tobacco-3482 dataset $\leftarrow \mathbf{D2}$
12. Repeat steps 5–10 for each instance in $\mathbf{D2}$
- # Step C: classification on RealDoc-16 dataset $\leftarrow \mathbf{D3}$
13. Repeat steps 5–10 for each instance in $\mathbf{D3}$

TABLE 2 | Statistics of datasets.

Dataset	Type	Total samples	Train split	Val split	Test split	No. of classes	Fewest samples per class	Most samples per class
RVL-CDIP	Secondary	399,743	319,754	39,995	39,995	16	24,951	25,000
Tobacco3482	Secondary	3482	—	—	3482	10	120	620
RealDoc-16	Primary	160	—	—	160	16	10	10

- b. **Tobacco3482:** Another dataset that is publicly available and has been used in this study is the Tobacco3482 dataset. The RVL-CDIP dataset contains all of the classes included in Tobacco3482, except for two, the “Note” and “Report” classes. “Note” and “Report” are only found in the Tobacco3482 dataset and not in the RVL-CDIP dataset. When compared to RVL-CDIP, Tobacco3482 evaluates the classification performance in low-resource and class-imbalanced conditions [77]. This dataset is, therefore, used to assess robustness and generalization under low-resource conditions.
- c. **RealDoc-16:** This study assembles a self-created dataset to assess the resilience of our suggested approaches in scenarios with restricted data. This dataset consists of 120 document images distributed across 16 classes, aligning with the class taxonomy of the RVL-CDIP dataset. The number of samples in each class is 10. This dataset was created to replicate real-world, low-resource situations and acts as an additional testbed to evaluate generalization performance beyond standard benchmark datasets. Some sample images from this dataset are given in Figure 4 below.

3.4 | Preprocessing of Datasets

Preprocessing of datasets is a very important step in DIC. It enhances the quality of both visual and textual modalities and assures that an optimal input is fed to the DL models while mitigating the challenges that can be posed by noise or irrelevant features. This study preprocessed both image and text data separately.

3.4.1 | Preprocessing of Images

The original document image size in the datasets is about 1000×750 pixels, but the vision-based DL model employed in this study, i.e., EfficientNetB3, expects an input size of 300×300 . All the document images were resized accordingly. For stable and efficient training, image pixels were normalized to the [0,1] range, and grayscale documents were converted into RGB format to comply with the input requirements of EfficientNetB3.

3.4.2 | Preprocessing of Extracted Text

Both of the proposed approaches in this study rely on the textual content present within the document images. For text extraction

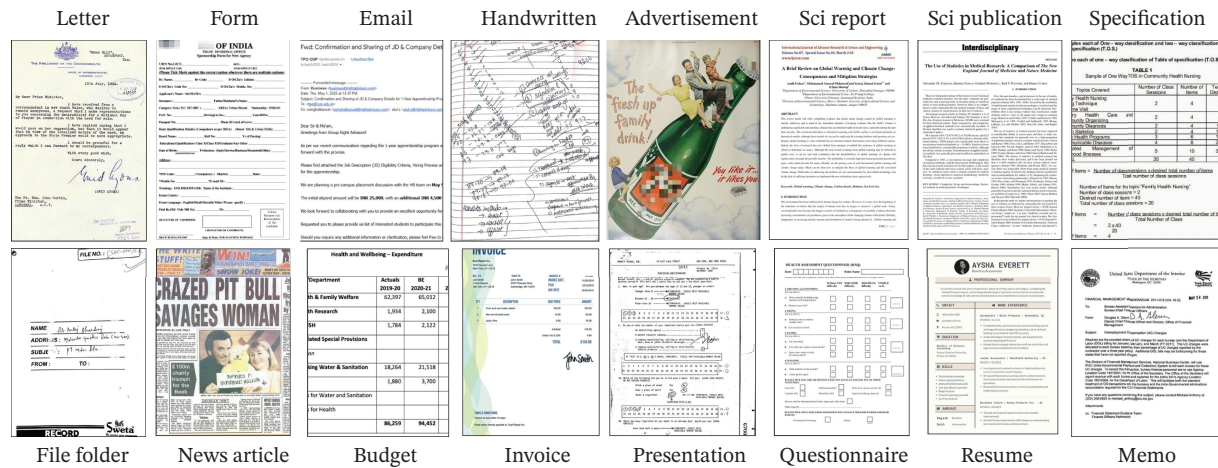


FIGURE 4 | Sample document images from RealDoc-16 dataset.

from document images, a commercial OCR engine served as the employed OCR engine. However, some document images in the dataset suffer from low resolution, and the OCR engine may fail to extract words accurately, resulting in missing, unclear, or incorrectly formed words, commonly known as OOV words. This problem was addressed by applying several preprocessing steps to the extracted text corpus which included the following: (a) transformation of all characters to lowercase, (b) tokenizing text into smaller tokens for model processing, (c) replacing multiple consecutive spaces with a single space, and (d) removing of digits, stopwords, special characters, and punctuation marks.

3.5 | Experimental Environment

All the experiments were carried out on an Ubuntu operating system (22.04.6 LTS) with an Intel Xeon(R) CPU E5-2620 v3 @2.40 GHz $\times$ 12, 125.7 GiB RAM, and an NVIDIA RTX A4000 GPU with 16 GB of VRAM utilizing CUDA v10.1. TensorFlow and Keras frameworks, along with Python 3.12.4, were used for developing the DL models.

3.6 | Training Details

Throughout this study, DL models were trained and fine-tuned at three stages: (i) fine-tuning “EfficientNetB3” for the visual stream, (ii) fine-tuning BERT_{base} for the textual stream, and (iii) fine-tuning the “BAAI/bge-small-en” embedding model for performing SCSR approach. The implementation details about the development of these DL models are outlined below.

3.6.1 | Fine-Tuning EfficientNetB3 for the Visual Stream

“EfficientNetB3” was fine-tuned using 319,754 document images from the RVL-CDIP training subset for 10 epochs with a batch size of 16. Adam optimizer at a learning rate of $1e-3$ was employed, and categorical cross-entropy was used as the loss function for this multiclass classification task. A dropout layer was incorporated to reduce overfitting, and the model’s performance was monitored using the RVL-CDIP validation split. Total time taken to fine-tune this DL model took about ~ 22.5 h, processing at a speed of ~ 3.5 batches per second. Figure 5 illustrates the model architecture used in this study.

3.6.2 | Fine-Tuning BERT_{base} for the Textual Stream

BERT_{base} was fine-tuned on 319,754 text instances extracted from the RVL-CDIP training split. Over 10 training epochs with a batch size of 16, Adam optimizer was used, and the learning rate, initially set to $5e-5$, was adaptively adjusted according to the validation performance. The dropout rate of attention and hidden layers was set to 0.3 to prevent overfitting, and model performance was monitored on the text extracted from the RVL-CDIP validation split. Total time taken to fine-tune this DL model took a total of about ~ 41 h, and each epoch took about ~ 4 h and 5 min.

3.6.3 | Fine-Tuning the BAAI/bge-small-en Embedding Model Using Supervised Contrastive Learning

The “BAAI/bge-small-en” embedding model was fine-tuned through supervised contrastive learning using an anchor-positive dataset, which was generated from the RVL-CDIP training split. This fine-tuning involved 4 epochs with a batch size of 16, 16 steps of gradient accumulation, and an AdamW optimizer with a learning rate of $2e-5$, incorporating cosine learning rate scheduling and a 10% warmup. To optimize the embedding space for semantic retrieval, the optimization objective combined multiple negatives ranking loss (MNRL) with Matryoshka loss function (MLF). This enables embeddings to capture multiple levels of granularities by drawing anchors closer to positive examples while pushing them away from multiple in-batch negatives in the embedding space.

3.7 | Evaluation Metrics

Experiments were performed to evaluate the performance of the proposed approaches on the RVL-CDIP test split and on the full Tobacco3482 and RealDoc-16 dataset. For RVL-CDIP and RealDoc-16, macroaveraged [78] accuracy, precision, recall, and F-1 score were reported, where equal importance is given to each class owing to their balanced distribution of instances throughout all the classes. In the case of Tobacco3482, which had an imbalanced class distribution, weighted-average precision, recall, and F-1 score were reported.

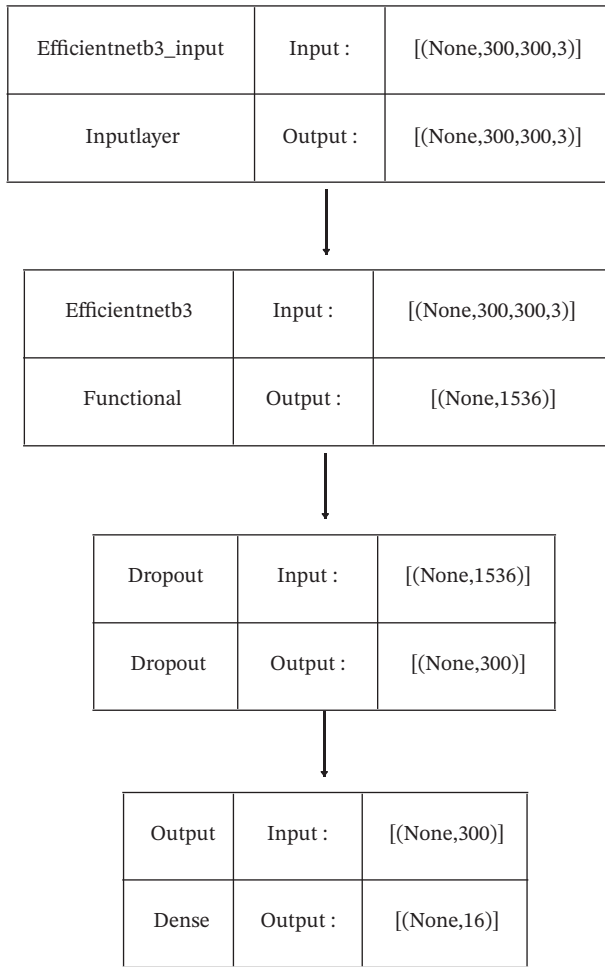


FIGURE 5 | Model architecture of EfficientNetB3 employed in this study.

3.8 | Comparison Approaches

To illustrate the effectiveness of the proposed approaches for DIC, comparisons were made against both unimodal baselines and three prior studies that employed multimodal fusion frameworks. For unimodal baseline models, BERT_{base} [31] and EfficientNetB3 [60] were selected since these models are employed within our framework for extracting textual and visual features, respectively. Among the prior studies, multimodal approaches proposed by Audbert et al. [3], Zingaro et al. [79], and Tyler et al. [44] were chosen for comparison. These studies were selected because they provide multimodal fusion frameworks for DIC and report results on at least one of the datasets used in our evaluation. Additional comparison studies were conducted for the SCSR approach with [24, 44] on the Tobacco3482 dataset to demonstrate the efficiency of a lightweight model free from dense, computationally heavy classification layers.

4 | Results and Discussion

This section presents the experimental results obtained by evaluating each of the proposed approaches separately. The performance of both approaches is compared against the employed unimodal baselines and three prior studies employing multimodal fusion framework.

4.1 | Performance Comparison With Unimodal Baselines

Figure 6 compares the accuracy results of unimodal baseline DL models employed in this study with the proposed approaches. BERT_{base} slightly outperforms EfficientNetB3 on both RVL-CDIP and Tobacco3482 datasets, suggesting that for much accurate classification, textual content present within a document image also plays an important role along with the visual layout of the image. However, the unimodal baselines achieved a lower accuracy score when compared to the proposed multimodal late-fusion approach on both RVL-CDIP and Tobacco3482 datasets, highlighting the benefits of combining textual and visual modalities for classifying document images.

Further, we visualized high-dimensional feature embedding spaces extracted by both unimodal baselines. Figure 7 illustrates the 2D uniform manifold approximation and projection (UMAP) of the features extracted from the employed unimodal DL models on the test set of RVL-CDIP. Compared with BERT_{base}, EfficientNetB3 produces well-separated clusters for document classes that are visually distinctive, such as *handwritten* and *advertisement*. In contrast, BERT_{base} makes more well-separated clusters for document classes with rich semantic content, such as *memo* and *scientific publication*.

Notably, unlike EfficientNetB3, the *handwritten* class is not well clustered by BERT_{base}, likely due to noise introduced during the OCR extraction process. Similarly, EfficientNetB3 struggled to separate *scientific report* documents, resulting in overlapping clusters, where BERT_{base} leverages the semantic information of this class to generate well-defined clusters. These complementary strengths and weaknesses highlight the need for the proposed multimodal framework, which utilizes both visual and textual modalities of an image to achieve more robust classification performance. A confusion matrix analysis for the baseline models and the proposed multimodal late-fusion framework are provided in the A.1 to validate the aforementioned modality-specific strengths across all document classes.

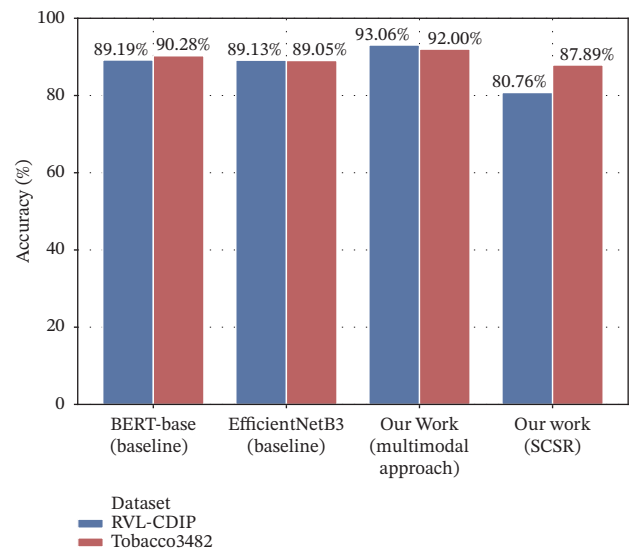


FIGURE 6 | Performance comparison of the proposed approaches with unimodal baselines.

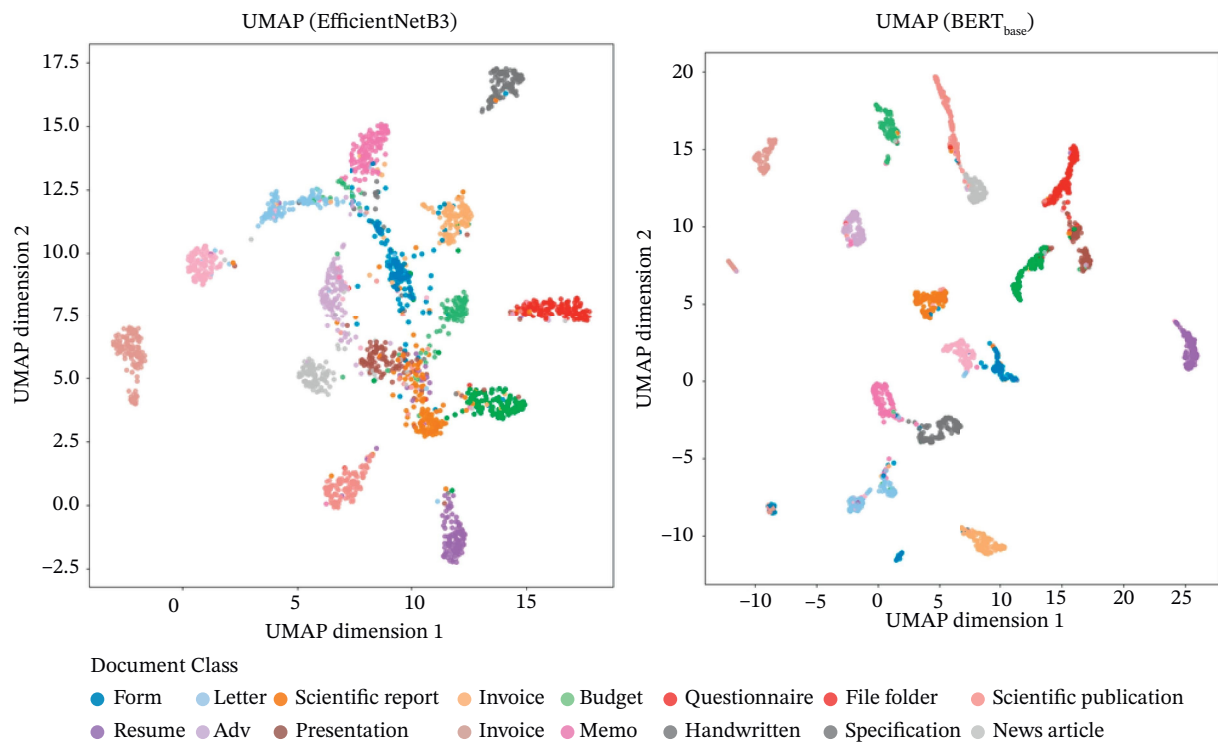


FIGURE 7 | 2-dimensional uniform manifold approximation and projection (UMAP) of the features extracted from EfficientNetB3 and BERT_{base} on the test set of RVLCDIP.

Further, the proposed alternative lightweight SCSR approach achieved promising results on the Tobacco3482 dataset, while struggled slightly on the RVL-CDIP test set. RVL-CDIP is a much larger dataset with high interclass similarity and low intraclass variations. In such cases, textual overlap can be common. Such datasets benefit more from heavy and computationally expensive DL models, which are trained with end-to-end supervision to learn fine-grained discriminative features. In contrast, the proposed SCSR approach lacks this level of fine-grained learning, which limits its performance on RVL-CDIP. Although this lightweight and efficient architecture does not employ any traditional DL models and remains competitive with the employed unimodal baselines, it provides a better lightweight alternative for lightweight DIC in resource-constrained environments.

4.1.1 | Attention-Guided Complementary Error Mitigation

By employing an attention-guided interpretability analysis, we visualize the error correction capability of the multimodal late-fusion framework on a randomly sampled document image from the RVL-CDIP test set. Table 3 visualizes the sample image where the ground truth is a “Questionnaire.” As illustrated by the Grad-CAM spatial attention heatmap, due to significant optical noise and formatting ambiguities, the visual stream, employing EfficientNetB3, incorrectly predicts it to be an “Email” by focusing on its spatial artifacts and ambiguous structure. Further, the textual stream employing BERT_{base} also incorrectly predicts the document to be a “Budget,” likely due to the OOV words introduced by inconsistent OCR extraction or numerical data. The token-level attention, extracted using integrated gradients, indicates that the DL model emphasized on such misleading textual cues. The multimodal late-fusion framework mitigates

the errors made by both unimodal models by leveraging complementary visual and textual modalities and predicts the correct document class. Similar attention-guided visualizations for a representative sample from each of the dataset’s classes have been plotted and are provided in Appendix A.2.

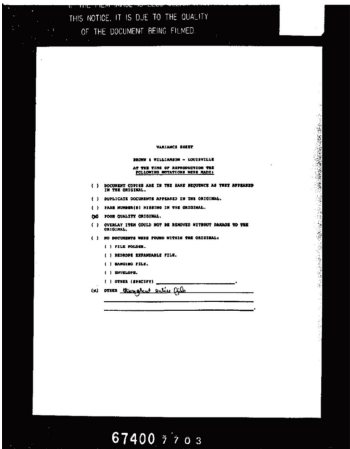
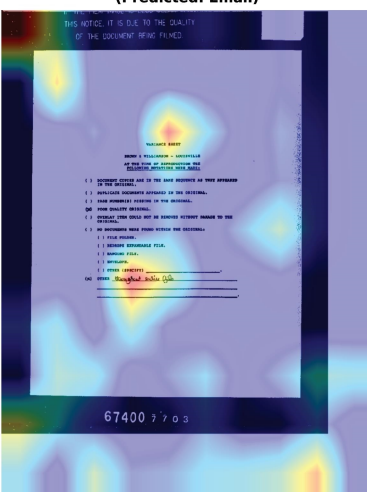
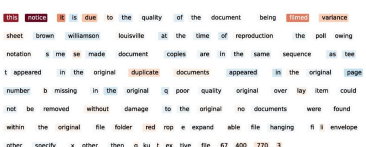
Figure 8 illustrates how the proposed multimodal late-fusion frameworks successfully mitigate the isolated peak errors of the employed individual modalities. Consequently, their shared secondary confidence in the true “Questionnaire” class is amplified, allowing it to emerge with the highest combined probability and resulting in correct classification.

4.2 | Performance Comparison With Prior Multimodal Frameworks

The results given in Table 4 highlight how both of the proposed approaches perform relative to three well-known existing multimodal methods for DIC.

Audebert et al. [3] employed a multimodal approach to classify the document images of RVL-CDIP and Tobacco3482 datasets, reporting accuracies of 90.6% and 87.8%, respectively, which improved upon their unimodal baselines of MobileNetV2 for visual stream and FastText embeddings for the textual stream. Their evaluation on both the datasets reported lower accuracy when compared with our proposed multimodal late-fusion approach, which was supported by stronger unimodal baselines for feature extraction. The proposed SCSR framework not only surpasses the unimodal baselines employed in [3] but also, despite its lightweight architecture and independence from DL, achieves performance comparable to both [44, 79] and the proposed multimodal framework.

TABLE 3 | Qualitative analysis of a representative RVL-CDIP test sample, showing the input document image, visual and textual attention regions, and the corresponding predictions of the visual, textual, and multimodal models.

Input document image	Grad-CAM attention (EfficientNetB3)	Token-level attention (BERT _{base})
Original Document 	Visual Focus (Predicted: Email) 	Textual Focus (Red = High Importance) (Predicted: Budget) 
True label: Questionnaire Visual stream: Email Textual stream: Budget Multimodal fusion: Questionnaire		

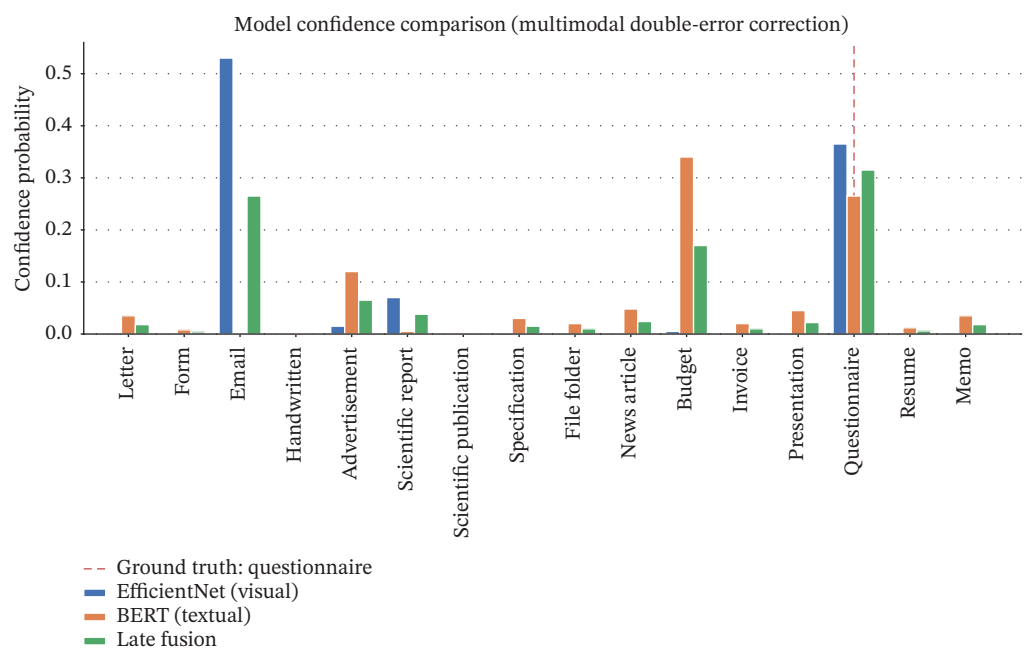


FIGURE 8 | Confidence probability distributions comparing peak unimodal misclassifications against the correct late-fusion prediction.

Zingaro et al. [79] introduced a multimodal approach using a side-tuning strategy. Their suggested multimodal approach reported an accuracy of 92.7 and 90.5, respectively, which was lower when compared to the proposed multimodal approach in this study on both the datasets. On the Tobacco3482 dataset, our unimodal textual baseline significantly outperforms their unimodal textual baselines, whereas on the RVL-CDIP dataset,

the unimodal results were not reported. This difference also arose because for the textual stream, they used FasText just like in [3], which does not capture richer features representations from the text as compared to BERTbase. The proposed SCSR approach, while lighter, performs similarly to the visual modality of their study where they employed MobileNetV2 and remains competitive to their suggested multimodal approach, confirming its

TABLE 4 | Comparison of the proposed approaches against prior studies employing multimodal approaches on RVL-CDIP and Tobacco3482 dataset.

Year	Study	Modality	RVL-CDIP				Tobacco3482			
			Acc	Macro			Acc	Weighted		
				F-1	Rec	Prec		F-1	Rec	Prec
2019	Audbert et al. [3]	T	74.6	—	—	—	73.8	—	—	—
		V	89.1	—	—	—	84.5	—	—	—
		T + V	90.6	—	—	—	87.8	—	—	—
2021	Zingaro et al. [79]	T	—	—	—	—	67.8	—	—	—
		V	—	—	—	—	88.0	—	—	—
		T + V	92.7	—	—	—	90.5	—	—	—
2019	Tyler et al. [44]	T	82.23	—	—	—	—	—	—	—
		V	90.24	—	—	—	—	—	—	—
		T + V	93.03	—	—	—	—	—	—	—
2025	Proposed multimodal approach	T	89.19	89.57	89.65	90.19	90.28	92.62	90.28	97.28
		V	89.13	89.12	89.15	89.23	89.05	91.58	89.05	96.15
		T + V	93.06	93.06	93.17	93.06	92.00	94.02	92.00	98.23
2025	Proposed SCSR approach	T	80.76	84.00	81.00	84.00	87.89	90.00	88.00	95.00

Note: T, V, and T + V denote text, vision, and multimodality, respectively. Bold values indicate the best-performing results across the compared approaches.

competitiveness as a lightweight alternative for classifying document images.

Tyler et al. [44] also proposed a multimodal framework for DIC. When compared, our textual unimodal baseline, which was employed using BERTbase, outperformed their textual baseline, which was using a BoW-based text model. For the visual baselines, our employed baseline EfficientNetV3 achieved comparable accuracy with greater efficiency. Their study employed AlexNet as the visual unimodal baseline, which is heavier with more parameters, due to which their visual baseline performs slightly better. Both multimodal frameworks achieved a similar accuracy on the RVL-CDIP dataset, with our approach performing slightly better. Our proposed framework also demonstrates strong cross-dataset generalization on Tobacco3482, which was not evaluated in their study. The proposed alternative lightweight SCSR approach, though lower in accuracy on the RVL-CDIP dataset, provides a lightweight and competitive alternative approach for DIC that does not depend on DL models.

4.3 | Per-Class Performance Comparison With Prior Studies

To better understand how well the proposed approaches generalize across different document types, per-class performance was evaluated on different document classes of the Tobacco3482 dataset. Among the aforementioned prior studies considered for comparison, i.e., [3, 44, 79], only Audbert et al. [3] and Zingaro et al. [79] reported per-class accuracies and only on the Tobacco3482 dataset. No per-class performance was provided for the RVL-CDIP dataset, while Dauphinee et al. [44] did not report per-class performance at all. Table 5 provides a detailed comparison of the per-class performance achieved by the proposed approaches.

The proposed multimodal fusion approach shows consistent improvements across almost all document classes when compared with prior studies. Our multimodal fusion approach achieves 98.41% on letter, 97.73% on forms, and 88.39% on sci-pub, representing significant gains over earlier approaches, which struggled with these classes. For classes like resume and memo, this framework achieves 100% and 99.68% accuracy,

TABLE 5 | Per-class performance of the proposed approaches against prior studies, evaluated on the Tobacco3482 dataset.

Study	Modality	Letter	Form	Email	Adv	Sci-pub	News	Resume	Memo
Audbert et al. [3]	T	58.8	77.0	29.5	93.3	64.4	63.6	60.7	49.7
	V	—	—	—	—	—	—	—	—
	T + V	93.2	88.7	99.1	94.8	66.7	95.5	95.3	93.0
Zingaro et al. [79]	T	71.0	76.0	96.0	60.0	57.0	67.0	97.0	79.0
	V	83.0	85.0	96.0	94.0	62.0	89.0	80.0	90.0
	T + V	86.0	88.0	98.0	93.0	68.0	90.0	96.0	90.0
Proposed multimodal approach	T	96.25	96.67	99.00	96.30	89.09	96.72	100.0	98.55
	V	98.21	96.29	99.50	97.69	80.37	89.19	98.32	97.87
	T + V	98.41	97.73	99.50	98.17	88.39	96.79	100.0	99.68
Proposed SCSR approach	SCSR	94.90	87.90	98.80	79.60	73.50	87.20	100.00	91.00

Note: T, V, and T + V denote text, vision, and multimodality, respectively. Bold values indicate the best-performing results across the compared approaches.

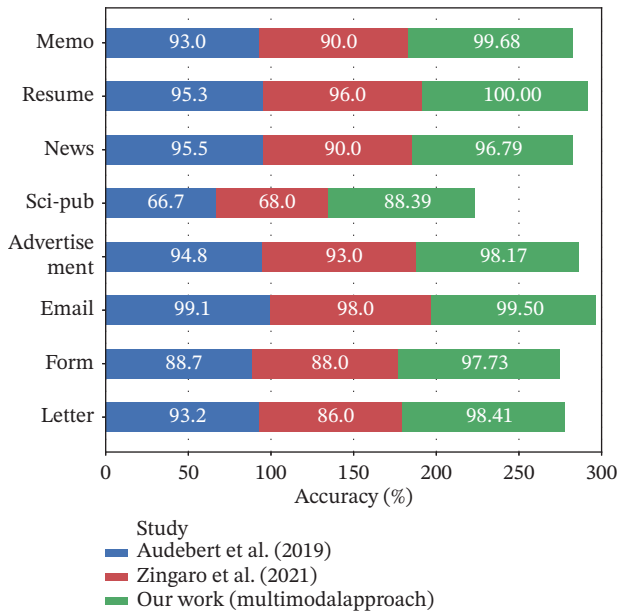


FIGURE 9 | Per-class performance comparison of the multimodal fusion framework against prior studies.

respectively, outperforming earlier approaches. Figure 9 further illustrates the class-level improvements achieved by the proposed multimodal framework. The proposed alternative lightweight SCSR approach also shows competitive performance, exceeding 90% accuracy on several classes, including 100% on resume and 91% on memo. This approach, in several cases, even surpasses the

TABLE 6 | Comparison of accuracy and parameter efficiency of the proposed SCSR approach.

Study	Accuracy (%)	Modality	Number of parameters
AlexNet-ELM [80]	83.24	I	62M
Zingaro et al. [79]	67.8	T	1.8M
Proposed SCSR approach	87.89	T	33.3M

Note: I and T denote image and text, respectively. Bold values indicate the best-performing results among the compared approaches.

TABLE 7 | Comparison of computational performance metrics between the proposed multimodal and SCSR approaches.

Approach	Total inference time (s)	Average latency per document (s)	Throughput (doc/s)
Multimodal fusion approach	555.2723	0.184	5.42
Proposed SCSR approach	547.1904	0.157	6.36

Note: Bold values indicate the best-performing results among the compared approaches.

TABLE 8 | Class-wise accuracy on RealDoc-16 dataset.

Class	Multimodal approach (%)	SCSR approach (%)
Letter	100.00	90.00
News article	100.00	80.00
Memo	72.73	63.60
Form	70.00	70.00
Resume	70.00	80.00
Scientific publication	70.00	90.00

unimodal baselines of prior studies. Although its performance is generally lower than the multimodal fusion framework, these results highlight the capability of dense semantic retrieval to capture class-relevant information without relying on any DL methods, making it an efficient approach for lightweight DIC.

4.4 | Efficiency and Performance Comparison of the Proposed SCSR Approach

To demonstrate the efficiency of the proposed lightweight alternative SCSR approach, we compared it to two different unimodal DL models with dense heavy classification layers. Despite having fewer parameters, our approach achieved higher accuracy on the Tobacco3482 dataset than DL models employed in [79, 80]. The results shown in Table 6 prove its efficiency as an alternative for deployment in low-resource settings.

Even though SCSR does not have the lowest number of parameters as compared to [79], it achieves a better accuracy-complexity trade-off, requiring substantially fewer parameters, i.e., 33.3M than AlexNet-ELM [80], i.e., 62M, and delivers higher classification performance. Additionally, computational performance metrics of the proposed SCSR approach were compared with the proposed multimodal approach. The SCSR approach exhibited faster inference performance, supporting its deployment in resource-constrained environments. The details of the computational performance metrics are given in Table 7.

4.5 | Performance Evaluation on RealDoc-16

Both of the proposed approaches were also evaluated on RealDoc-16, a separately gathered document image dataset, aligning to the 16 classes of the RVL-CDIP dataset. Despite a smaller sample size for each class and a significant variability in the document images, both of the proposed approaches showed a promising performance across multiple classes. Table 8 below illustrates the experimental results for this self-curated dataset.

5 | Conclusion and Future Work

To address the limitations posed by traditional unimodal approaches and the increasing demand for lightweight and resource-efficient models for DIC in real-world and resource-constrained environments, this study proposed two distinct, independently developed DIC frameworks. The multimodal late fusion framework was proposed, which combines the visual and textual modalities of the document image at decision-level to achieve maximum possible classification accuracy. This

approach outperformed the unimodal baselines and prior studies which employed multimodal frameworks with heavy complex architectures. The approach achieved accuracies of 93.06% on RVL-CDIP and 92.00% on Tobacco3482, while also showing significant cross-domain generalization on RealDoc-16. Another independent lightweight approach was proposed that leverages supervised contrastive learning and semantic retrieval to deliver competitive performance with significantly lower computational requirements. The performance of this lightweight approach remained competitive, especially on Tobacco3482 (87.89%) and RealDoc-16. For RealDoc-16, it even outperformed the multimodal fusion framework on several categories, highlighting its potential as a lightweight yet effective solution for classifying document images in real-world and resource-constrained environments. Future research could investigate more robust representation learning frameworks for multilingual document classification in low-resource settings with severe OCR degradation. To facilitate deployment in resource-constrained environments, parameter-efficient fine-tuning (PEFT) can be integrated to reduce training memory requirements. Adaptive modality weighting or large-scale multimodal foundational models with confidence-aware fusion could be explored, replacing the current unweighted average ensemble for multimodal fusion. Despite showing competitive performance, the limitations of SCSR on the RVL-CDIP dataset, caused by high interclass similarity between classes and textual overlap, could be mitigated by including lightweight image embeddings into the retrieval database.

Author Contributions

All authors agreed on the content of the study. Anshul Kashyap was responsible for writing the main manuscript text, data curation, methodology, and implementation. Satwinder Singh contributed to conceptualization, supervision, and served as the corresponding author. Meenakshi Mittal contributed to the editing of the manuscript.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

RVL-CDIP: This is a publicly available dataset, and no explicit permission is required for its use. The dataset can be accessed online on <https://adamharley.com/rvl-cdip/>, and its license information can be found at <https://industrydocuments.ucsf.edu/about-idl/copyright/>.

Tobacco3482: This is also a publicly available dataset, and no explicit permission is required for its use. The dataset can be accessed online on <https://huggingface.co/datasets/maveriq/tobacco3482>.

RealDoc-16: The data will be provided on a reasonable request.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section. **Supporting Information.** Building upon Section 4.1 Performance Comparison with Unimodal Baselines and Section 4.1.1 Attention-Guided Complementary Error Mitigation, the supporting information for this manuscript is provided as the Appendix, which is divided into two sections. Section A.1 includes a detailed confusion matrix analysis comparing the performance of the employed unimodal baselines against the proposed multimodal late-fusion framework on the test set of RVL-CDIP. To further demonstrate the complementary error mitigation of the proposed multimodal late-fusion approach, Section A.2 presents extended qualitative visualizations for each document class, including visual spatial attention heatmaps generated via Grad-CAM and token-level attribution maps extracted using integrated gradients.